



**NANYANG
TECHNOLOGICAL
UNIVERSITY**
SINGAPORE

Consult 3: CA6121 AI in Project Management

Group Project Formulation, Schedule development and to proceed to next steps

Presented by Dr Ravish Maqsood





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Project Topic and Approach

AI in Project Management



Group topic submission

- Stay in touch with your group mates.
- Get to know each other.
- Explore the topics in depth for the group assignment do you wish to work on.
- Come up with **proposed topic** for your group.
- **Submit your group topic** one day before the next consultation class.

<https://forms.cloud.microsoft/r/B8zBtFa6Nb>



[CA6121 NTU Group Assignment Topic selection Mar2026 – Fill out form](https://forms.cloud.microsoft/r/B8zBtFa6Nb)



Planning Concept

Project Planning & Scheduling



Planning

A plan is used to:

- Align expectations of all parties involved
- Minimize workflow time of a project
- Minimize or avoid conflict
- Manage changes effectively
- Resource allocation
- Many more...



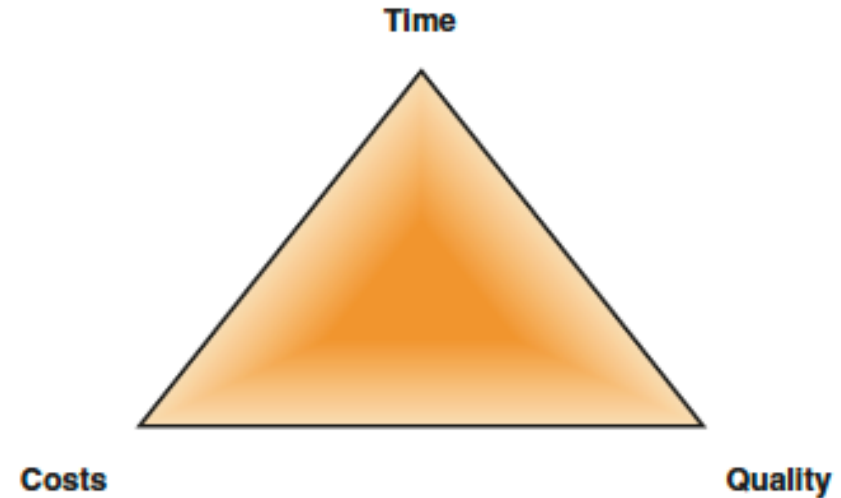
Planning helps to answer:

- **What?**
 - Identify and describe the subprojects, work packages and tasks; The result is the work breakdown structure.
- **Until when?**
 - Describe the timeline; The result is the time schedule and milestone plan.
- **Who/how many?**
 - Show which forces are to be used with which qualification, which personnel and material resources are needed. This is documented in the resource and cost plan. The project organization describes the type of cooperation.
- **Conditions?**
 - Specify the technical and spatial requirements, list the known restrictions. This is noted directly in the project plan.



Planning challenge to balance following three aspects

- **Quality**
- **Cost**
- **Time**





Project Planning and Scheduling

Project Planning & Scheduling



Key Processes

- 1. Define Activities**
- 2. Sequence Activities**
- 3. Estimate Activity Duration**
- 4. Develop Schedule**



Gantt Chart

Project Planning & Scheduling



Gantt Chart

A Gantt chart,
commonly used in project management
tool to show activities (tasks or events)
displayed against time.



Gantt Chart

AI Tool can help

Gantt chart allows to see:

- Various activities
- Start and finish of each activity
- Duration of each activity
- Activities overlap with other activities
- Critical path activities and Float
- The start and end date of the whole project



Project Topic and Approach

AI in Project Management



AI in Project Management:

- Criteria to select a project for group assignment:
 - Can have either or all phases of project covered in the lecture
 - Initiation,
 - planning,
 - executing,
 - monitoring & control,
 - close-out.
- Must draw a workflow of the current situation and proposed improvement.
- **Improve process** for a project by implementing AI tool of your choice. (Example: collating of notes of meeting, preparation of EVM, sorting the team profiles, etc.).

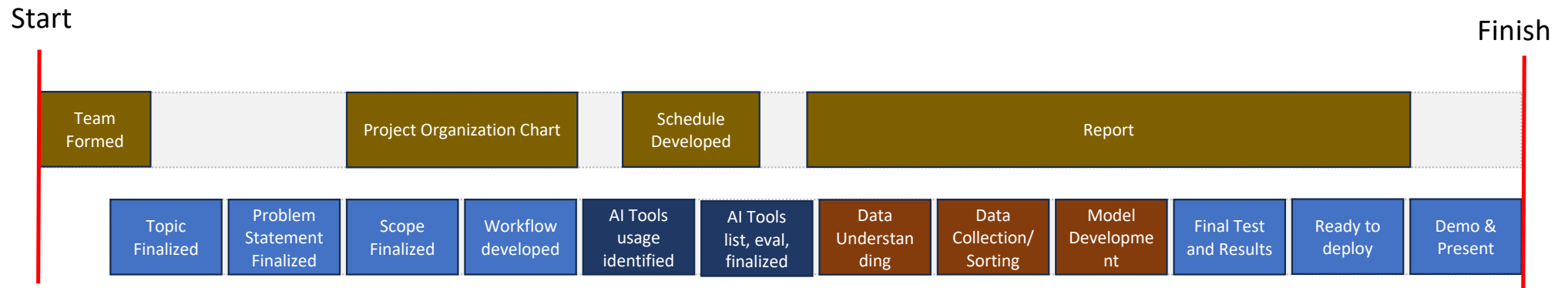


Your Group Project Progress?

AI in Project Management



Group Project Approach





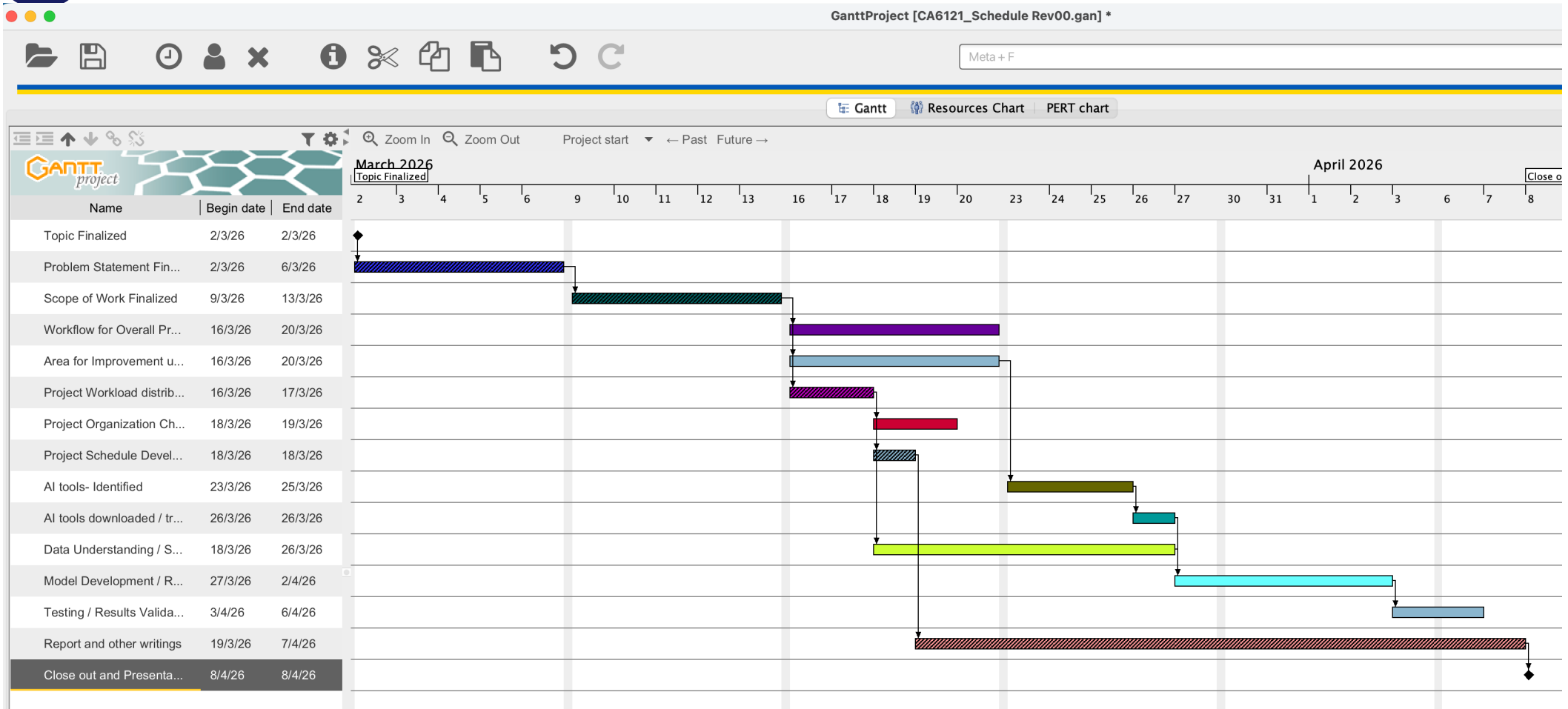
Project Steps & Planning- Gantt Chart

Project Execution Steps: For Reference only					
S.No	STEPS	Resources	Involvement	Duration	Predecessor
1	Topic Finalised	1,2,3,4,5 (All Team)	20% time of each person	1	Nil
2	Problem statement finalised	1,2,3,4,5 (All Team)	20% time of each person	5	Topic Finalised
3	Scope of work finalised	1,2,3,4,5 (All Team)	20% time of each person	5	Problem statement finalised
4	Workflow for the overall project developed	1,2,3,4,5 (All Team)	20% time of each person	5	Scope of work finalised
5	Areas for improvement using AI tools in workflow identified	1,2,3,4,5 (All Team)	20% time of each person	5	Scope of work finalised
6	Workload distributed among the team	1 (Project Manager)	20% time	2	Scope of work finalised
7	Project Organization chart is developed	1 (Project Manager)	20% time	2	Workload distributed among the team
8	Project Schedule Developed for the overall project	4 (Planning / Sch / Cost Expert)	20% time	1	Workload distributed among the team Project Organization chart is developed
9	AI Tools to be used identified, listed, evaluated, and finalized	2,3 (Data and AI Tools Expert)	20% time of each person	3	Areas for improvement using AI tools in workflow identified
10	AI Tools downloaded / installed / subscribed (free version).	3 (AI tools Expert)	10% time	1	AI Tools to be used identified, listed, evaluated, and finalized
11	Data understanding, Collection, Sorting completed for the project	1,2,3,5 (All team except Planning/Sch/cost)	10% time Project Manager 20% time Data Expert 20% time AI Tools expert 10% time Testing/ QA/ Deployment expert	7	Workload distributed among the team
12	Model development / Run / Testing Complete to use the data	1,2,3,5 (All team except Planning/Sch/cost)	20% time of each person	5	AI Tools downloaded / installed / subscribed Data understanding, Collection, Sorting completed for the project
13	Testing / Results are validated / Ready to Deploy	2,3,5 (Data & AI Tool Expert, Testing Expert)	100% time Data Expert 10% time AI Tools expert 20% time Testing/ QA/ Deployment expert	2	Model development / Run / Testing Complete to use the data
14	Report Writing : Benefits/ Challenges / Lessons for these AI tools on this project identified and recorded	1,2,3,4,5 (All Team)	20% time Project Manager 10% time Data Expert 10% time AI Tools exper 20% time Planning / Sch / Cost Expertt 10% time Testing/ QA/ Deployment expert	14	Project Schedule Developed for the overall project

- 1 Project Manager
- 2 Data Expert
- 3 AI Tools Expert
- 4 Planning / Sch / Cost Expert
- 5 Testing/ QA/ Deployment



Project Steps & Planning- Gantt Chart

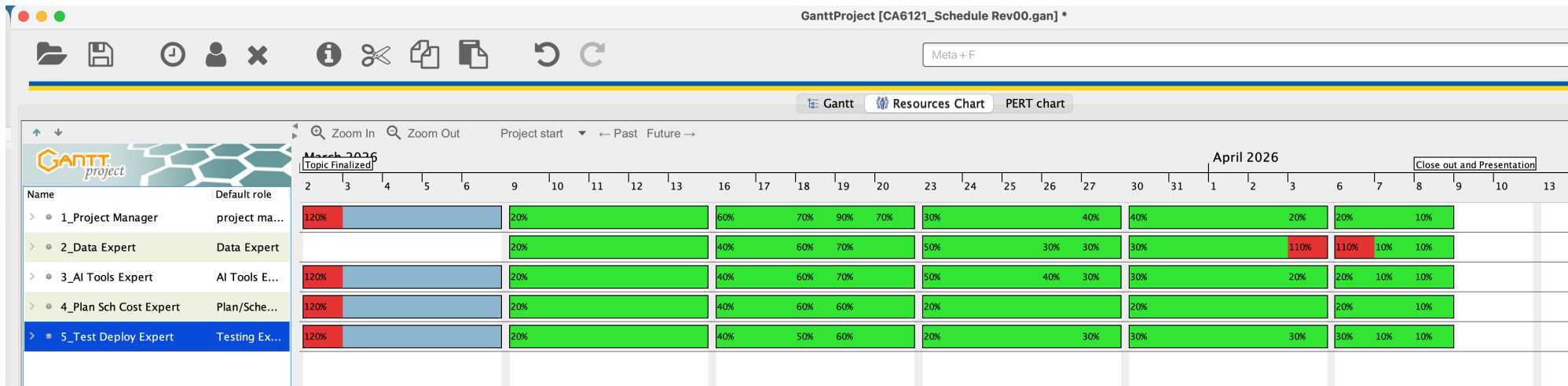


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Project Steps & Planning- Gantt Chart

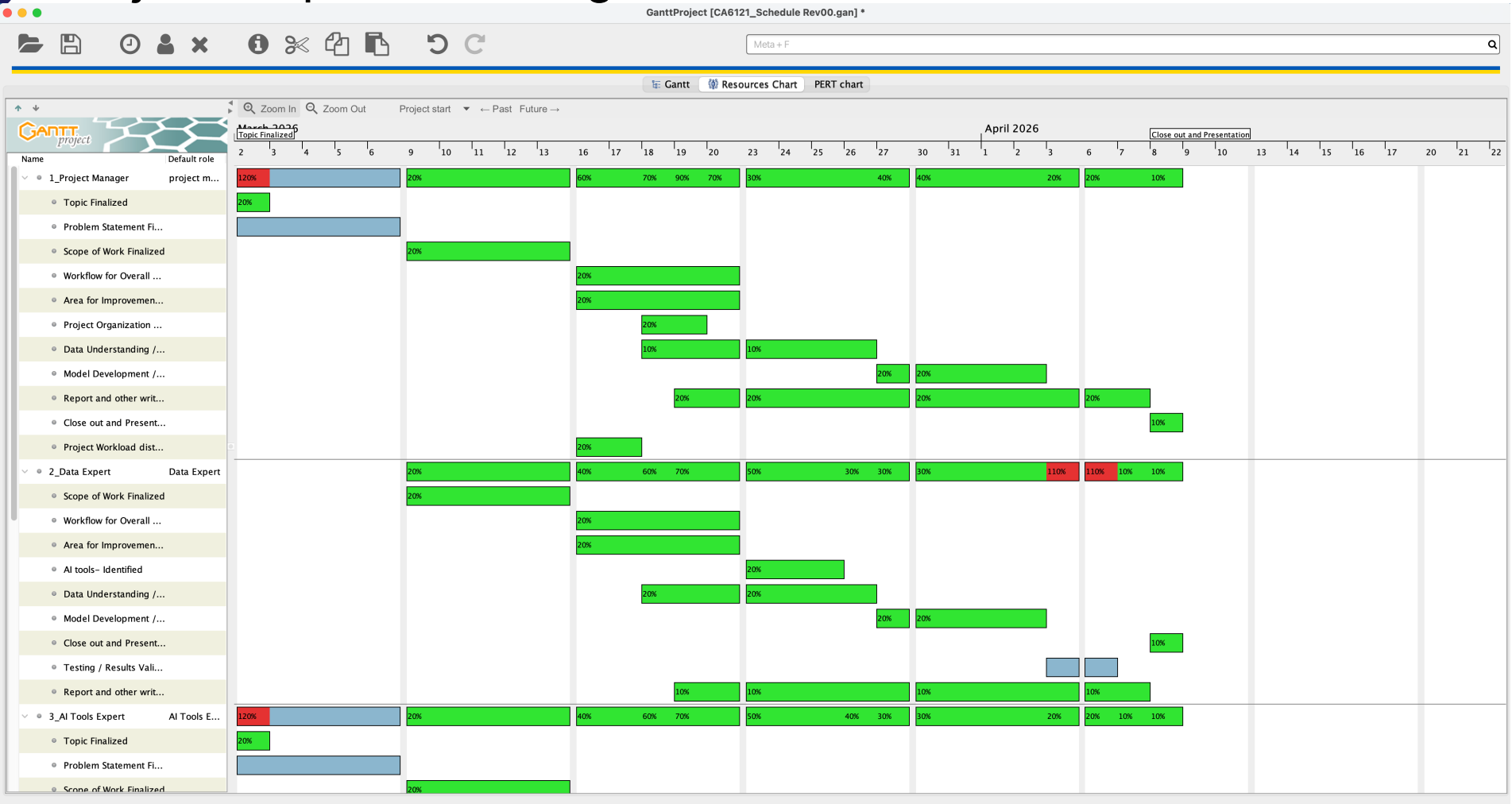


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Project Steps & Planning- Gantt Chart





Project Planning and Schedule



Download GanttProject

free Gantt chart for small and medium businesses

GanttProject 3.3

released on 15 Jan, 2024

License

short summary

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Project Planning and Schedule

Download GanttProject 3.3

This is the Release build of GanttProject 3.3.

Windows

Both EXE and MSI installers are executable programs and both are bundled with Java Runtime. You don't need to download anything else.

macOS

We recommend using the EXE installer, however, system administrators may want to choose the MSI installer to automate bulk installation job.

Linux

ZIP

GanttProject 3.3.3316 for Windows

Built on 30 Jan, 2025

License

[Download EXE installer \[130 Mb\]](#)

- GanttProject 3.3.3316 will be installed side by side with the previous version, if any.
- The MD5 sum of the installer file: [9505f15f583b155c52c6f50aa225e263](#)

[Download MSI installer \[132 Mb\]](#)

- GanttProject 3.3.3316 will be installed side by side with the previous version, if any.
- There is a [silent install option](#) for system administrators who need to perform bulk installations on user computers.
- The result of antivirus tests: [no viruses detected by VirusTotal](#)
- The MD5 sum of the installer file: [65d8eb2653878309c4b809c1559268a9](#)

<https://www.ganttproject.biz/download>



Your Group Project Progress?

AI in Project Management



Your Group Project Progress?

Recap from Tutorial 02

1. Teamwork is important.
2. Appoint a group lead who will act as project manager to coordinate this group assignment activities.
3. Group assignment topic is reviewed and agreed by all team members?
4. Problem Statement development status?
5. Have you finalized the scope?
6. Where are you on the workflow development for the overall project?



Topic understanding and to set a path to work on.

Recap from Tutorial 02

- Understand your group project topic.
(Use tutorial 01 class exercise as reference).
- What **type of project and which industry workflow** your project belongs to?
(Suggestion: IT, Construction, Building, Manufacturing, Aerospace, etc.?)
- Explain **current problem in this workflow** where your group is trying to solve by implementing AI tools in Project Management Process?
(Example: Notes of meeting compilation, review, analysis, draft email to be sent to the attendees).
- What is your **proposal to solve** the identified problem statement?
(Example: AI tool implementation to collate meeting transcript and compile into notes of meeting, perform review, analysis, and prepare draft email to be sent to the attendees)
- Draw a **workflow or block diagram or write down the steps** you will take to solve.
- Once topic is finalized, ask again: **is your team confident to work on this project topic or wish to revise or fine-tune?**

AI Tool can help

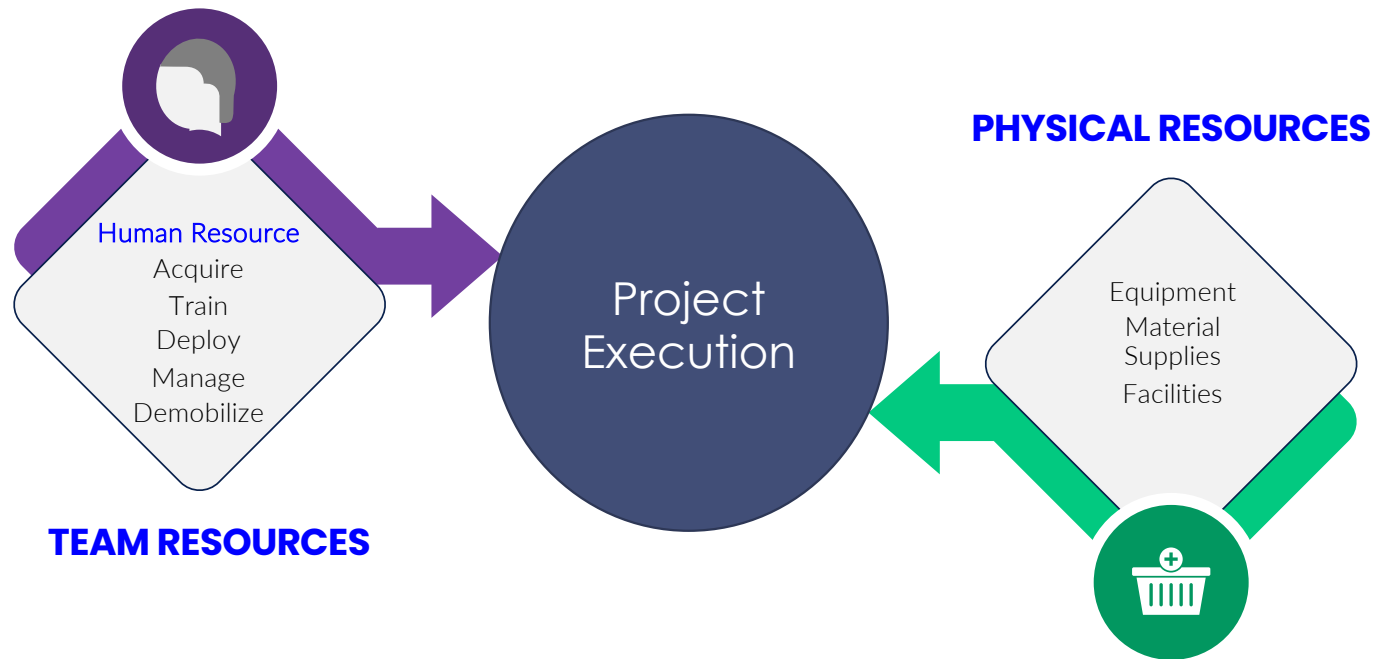


Project Resources

Project Resource Management



Type of Resources





Group Project Resources

- List down the resources on your project.
- Prepare physical resources list.
- Prepare team (human) resources list.
- Any challenges foreseen in your project?



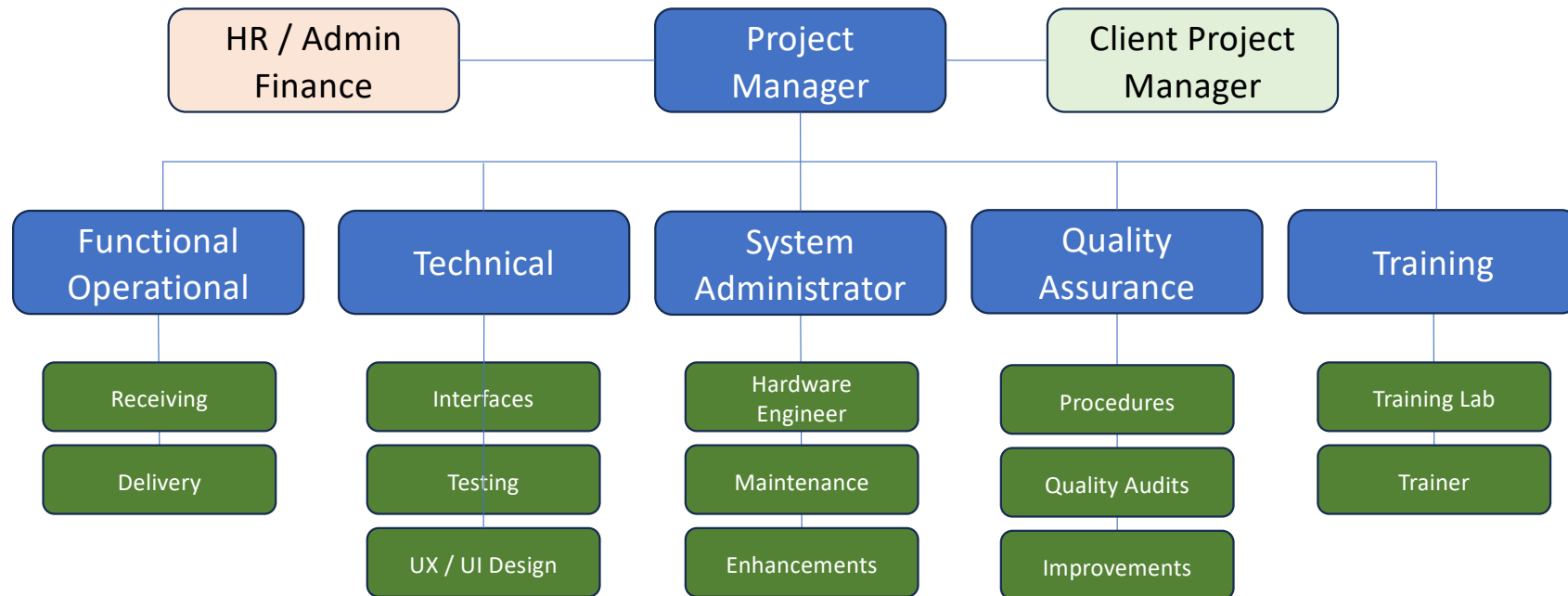
Group Project Organization

Project Resource Management



Group Project Organization

Sample:
Update as per your project requirements





Roles & Responsibilities

Project Resource Management



Project Roles and Responsibilities

Roles and Responsibilities are defined for all people on the project:

- Role
- Experience and competencies
- Skills and training



What to do before next consultation?



Group topic submission

- **Be prepared to discuss the progress of your group project** before the next consultation class.
- Be familiar with **Project Cost Management** terminologies.



What to do before next tutorial class?



Tutorial Classes

- Ensure you have **watched the relevant lecture(s)**.
- Familiarize yourself with
 - **Planning and Scheduling activities in a project**
 - **Gantt Chart**
 - **Critical Path Method (CPM)**
 - **Gantt Project Software or**
 - **any other software of your choice used for planning and scheduling.**
- Please **bring your laptop** to participate in class exercises/ activities.



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End of class



Batam, Indonesia

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